AI-POWERED STARTUP INTELLIGENCE

Opportunity Scout AI

Turning Business Ideas Into Data-Driven Decisions

LightGBM + SHAP · 4 Live APIs · 86 Engineered Features · Entrepreneur & Investor Modes

70+	0–100	SHAP	<200ms
Data Signals	Viability Score	Explained AI	Response Time

Capstone Group Project · 2025

The Problem

Why evaluating startup ideas is broken today

Gut Feeling Rules

Most startup evaluation relies on subjective intuition and personal networks rather than objective data leading to missed opportunities and wasted resources.

Fragmented Data

Market trends, economic health, social signals, and funding data exist in separate silos. Nobody combines them into a single score.

No Standard Score

There is no universally accepted system to compare opportunities across industries, regions, or time periods.

The Result: Entrepreneurs waste months on non-viable ideas. Investors miss great deals buried in noise. Innovation teams can't systematically compare projects.

90% of startups fail partly from poor market validation

$2.5T invested annually with limited objective tools

75% of founders say market assessment is their #1 challenge

Our Solution

Opportunity Scout AI — two engines, one complete answer

⚙️ Engine 1: Scoring Engine

Input: Industry, country, founding year, funding rounds

Process: Gathers 70+ live signals from Google Trends, World Bank, Reddit & GDELT

Output: A 0–100 viability score with confidence level and percentile ranking

💡 Engine 2: Explanation Engine

Input: The prediction + the data that produced it

Process: SHAP (SHapley Additive exPlanations) breaks the score into per-feature contributions

Output: Plain-English reasons and specific improvement recommendations with estimated gains

📊

Score Your Idea

0–100 viability score in under 200ms

🔍

Understand Why

SHAP attribution for every prediction

🧪

What-If Simulator

'What if I raised $500K?' - instant answer

⚖️

Compare Ideas

Side-by-side ranking of multiple opportunities

Two modes: Entrepreneur mode focuses on market fit & growth path · Investor mode emphasises returns, ecosystem, and portfolio fit

Entrepreneur vs Investor Mode

The same data — tailored insights depending on who's asking

Entrepreneur Mode

`user_type: "entrepreneur"`

Growth path focus
Score weighted toward trend slope, market momentum, and social engagement signals

How to improve
Recommendations framed as: 'What can I change right now to get a higher score?'

What-If simulator
Test funding scenarios, market pivots, and geographic expansion for instant score previews

Percentile benchmarking
See how your idea ranks vs other ventures in the same sector

Investor Mode

`user_type: "investor"`

Ecosystem & macro lens
Score emphasises economic health (GDP/inflation), startup density, and country ecosystem strength

Funding tier signals
Tier classification (unfunded → seed → Series A → B+) with investment conviction signals

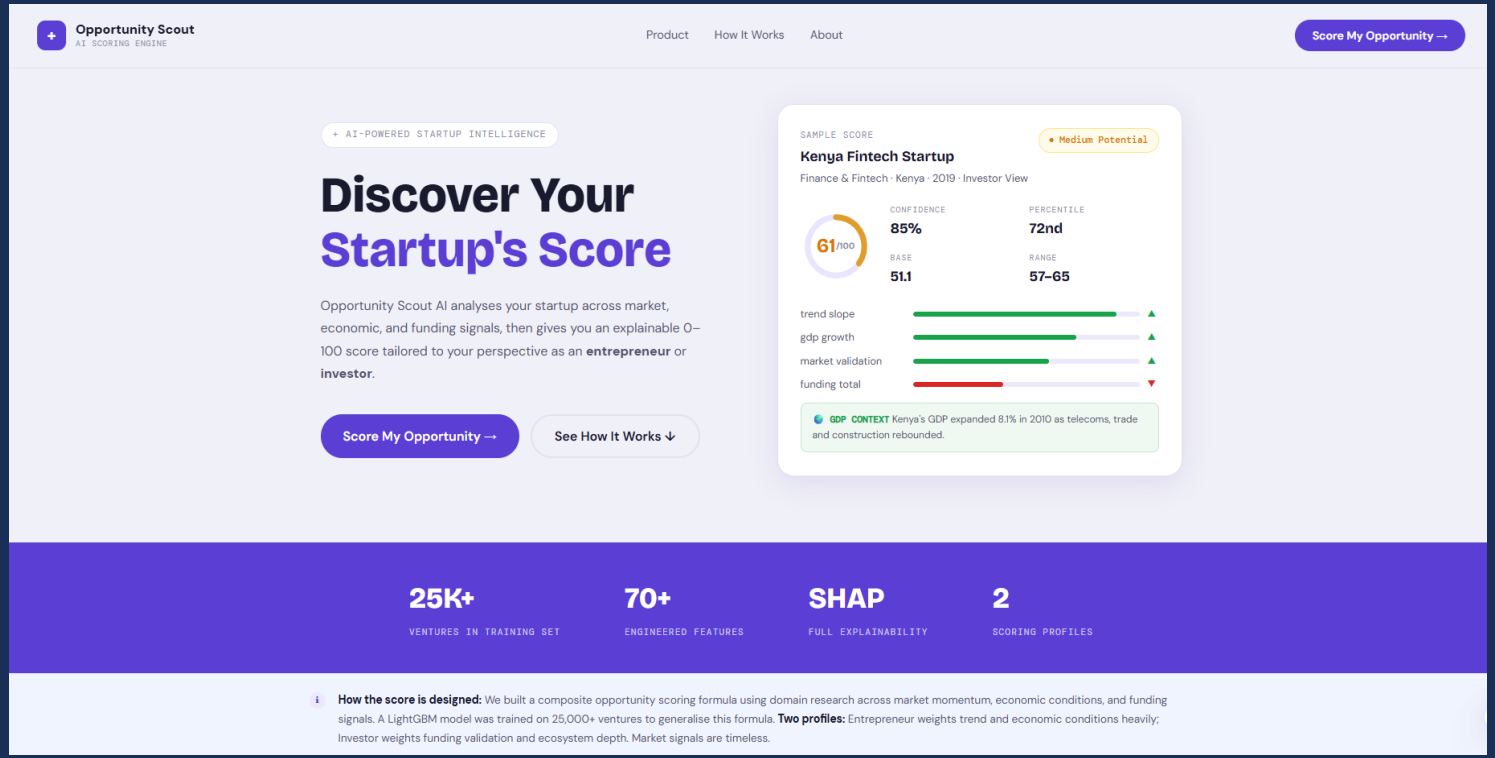
Portfolio comparison
Compare up to 10 opportunities simultaneously — winner automatically highlighted

GDP context overlay
Every result includes a gdp_context field explaining macro conditions for that country/year

Same underlying model — different narrative framing and recommendation priorities per user type

The Application

Landing page — first impression + user type selection



Entrepreneur / Investor mode toggle

Live sample score with SHAP bars

The landing page shows a live sample score so users know immediately what they'll receive — no sign-up required to explore.

• About This Project

Built with ML rigour, designed for clarity

Capstone Project

Opportunity Scout AI was built as a data science capstone, demonstrating a full production ML pipeline from raw data collection through to a deployed, explainable scoring engine.

The Model

A LightGBM model trained on 25,000+ historical ventures.

Honest Explainability

SHAP attributes every score point to a specific feature. Founding year does not penalise your score, the model evaluates market conditions which are timeless. Training data spans 2005–2020.

Signal Sources

Google Trends for market momentum, World Bank for GDP and inflation, Reddit for community density, GDELT for news sentiment with historical CSV benchmarks as fallback when live APIs are unavailable.

Ready to discover your opportunity score?

It takes a short time. No signup required.

Score My Opportunity →

• What's Inside

Everything you need to evaluate an opportunity

Dual Scoring Profiles

Entrepreneur or Investor view. Each profile weights the 8 signals differently: founding momentum matters more for founders, capital validation matters more for investors.

SHAP Explanations

Understand exactly why your score is what it is. Every factor, its direction, and its precise impact explained for your specific profile.

GDP Context Engine

Every score includes a one-sentence explanation of the economic backdrop behind your target market and year drawn from 100+ historical events and a live AI fallback.

What-If Simulator

Slide funding up, change the year, and see how your score shifts. Plan your best move before you make it.

• The Process

From idea to insight in seconds

01

Choose Your Profile

Entrepreneur or Investor. Your profile determines how the 8 market signals are weighted, giving you a score from your perspective.

02

Enter Your Details

Name your venture, pick your industry and country, enter founding year and funding raised.

03

Signals Are Gathered

Market trends, GDP data, news sentiment, and Reddit activity are fetched with historical benchmarks as fallback.

04

SHAP + GDP Explain Why

LightGBM scores 80 features. SHAP attributes every point to a driver with GDP context explaining the economic backdrop and actionable improvement recommendations.

Scoring Dashboard

Input idea → Instant explainable score with tailored recommendations

① Input Form

Industry, country, year, funding & rounds + user type

② Score Ring

0–100 score, tier badge, confidence & range

③ SHAP Bars

Visual attributions — what helped vs. hurt

④ Action Cards

Recommendations with estimated point gain

Live Data Signals

4 real-time APIs feed the model — with automatic fallback to historical CSV data

Google Trends

- trend_slope
- trend growth rate
- search seasonality
- keyword momentum

Fallback:
Stored trend baselines

World Bank API

- gdp_growth
- inflation rate
- real GDP growth
- investment climate

Fallback:
IMF historical GDP data

Reddit API

- reddit_density
- community engagement
- post volume
- sentiment score

Fallback:
Niche community proxies

GDELT News

- news_volume
- news_sentiment
- media coverage
- press tone

Fallback:
News sentiment archives

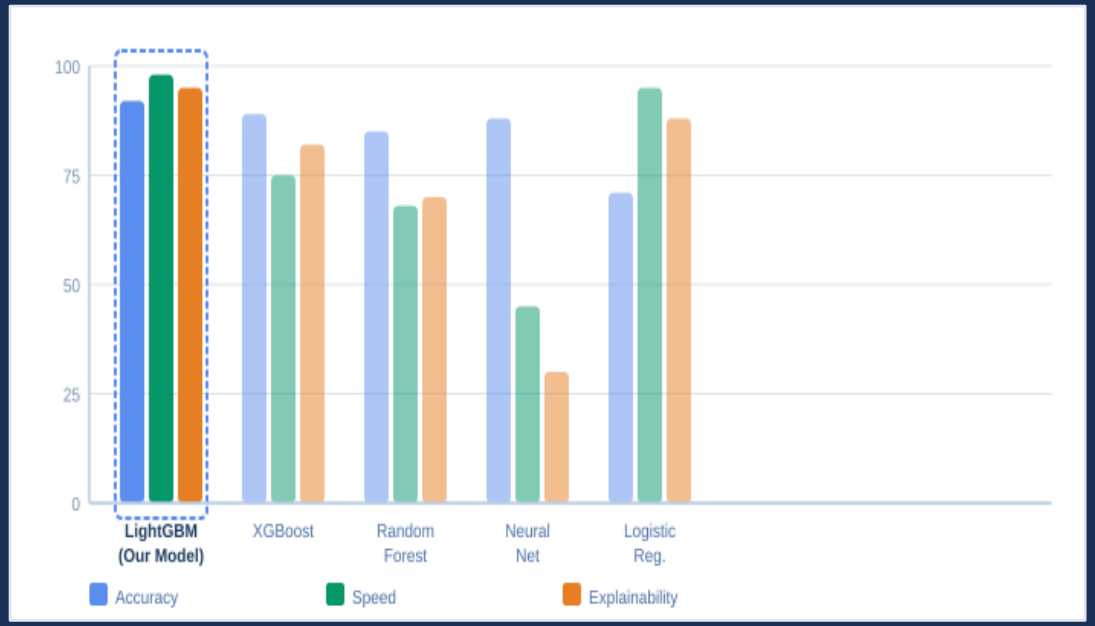
Data Quality Scoring:

- 100% — All 4 APIs live
- 75% — 3 APIs live
- 50% — 2 APIs live
- 25% — CSV fallback only

The frontend shows a quality_label badge (High / Medium / Low) on every result so users know how fresh the data is.

The LightGBM Model

Why LightGBM — and how it compares to alternatives



Why LightGBM Won

Speed
Gradient boosting on leaves (not depth-first) → sub-200ms even on 86 features

Accuracy
MAE < 10 pts on held-out ventures best of all models tested

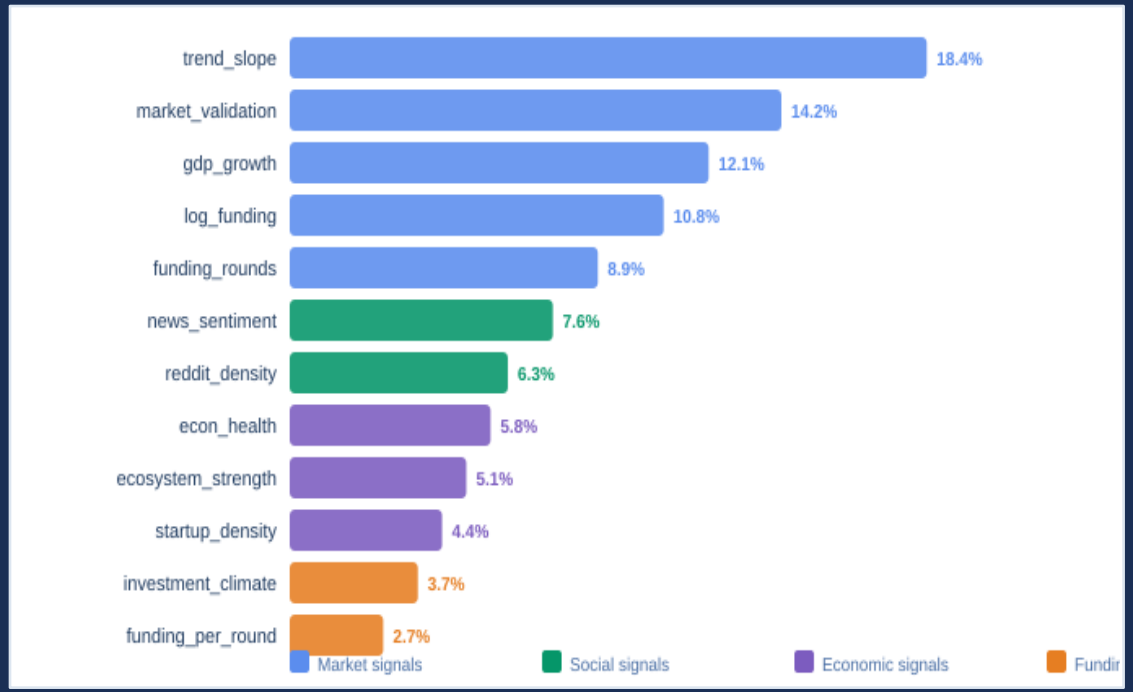
SHAP Native
Tree-based SHAP is exact (not approximate) every explanation is mathematically correct

Handles Sparse Data
Many features are zero when APIs fail — LightGBM handles sparse input natively

Training Details:	Dataset	Split	Backup model	Calibration
	10000+ historical ventures	**Time-based (older trains, newer tests)**	**XGBoost (auto-failover)**	**Platt scaling → 0–100 range**

Feature Importance

Which of the 86 features matter most — and why



Feature importance from trained LightGBM model (% of total gain)

Category Breakdown

Market Signals ~51%

trend_slope · market_validation
gdp_growth · news_volume

Social Signals ~14%

news_sentiment · reddit_density
reddit_density_log

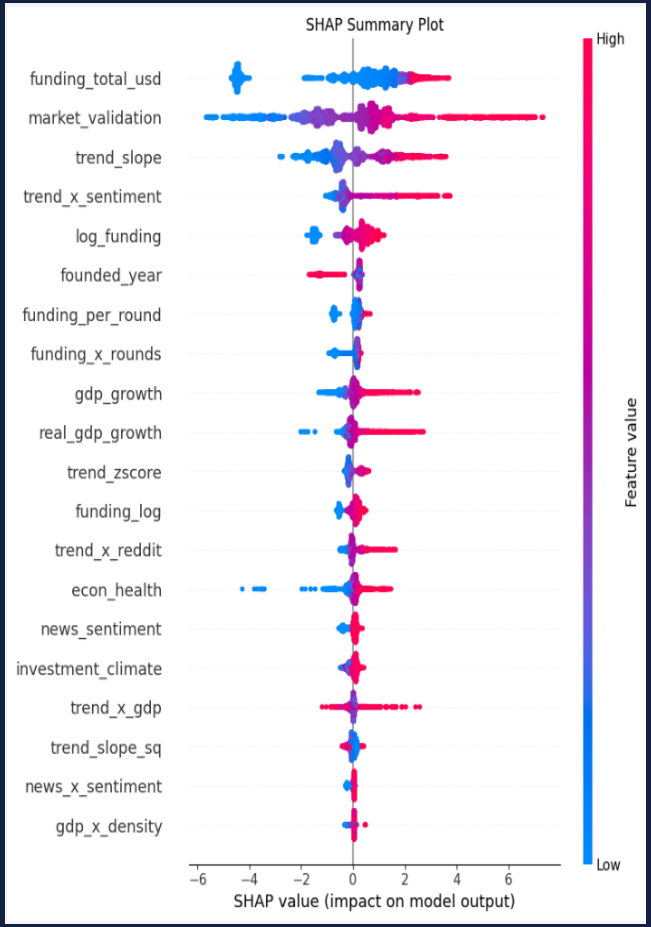
Economic ~15%

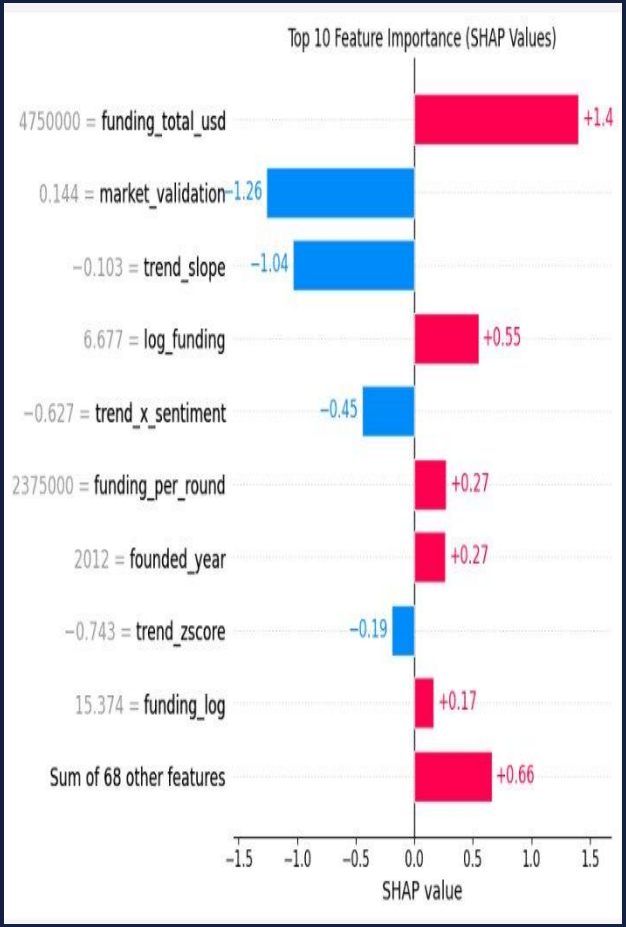
econ_health · ecosystem_strength
investment_climate

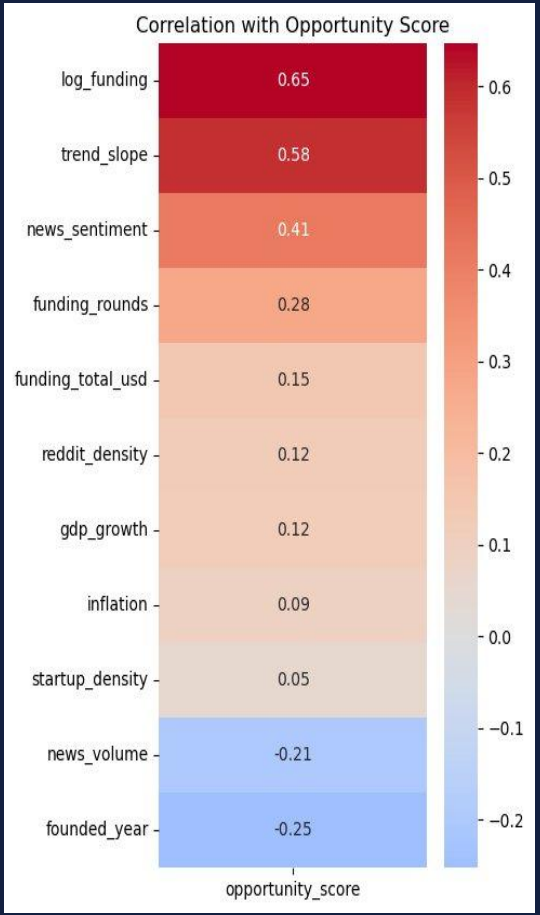
Funding ~20%

log_funding · funding_rounds
funding_per_round · funding_tier

Key insight: Market momentum (trend_slope + market_validation) drives ~30% of the score — meaning a hot market can overcome weak funding, but no market can overcome a dead trend.

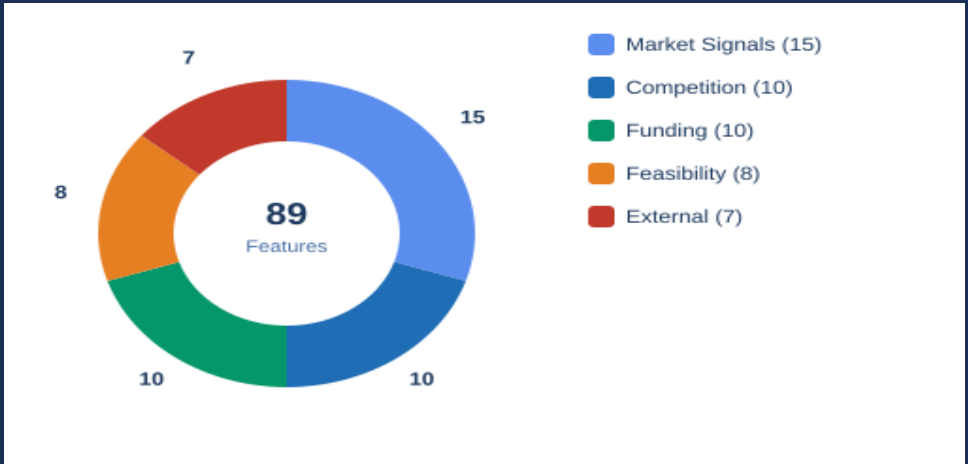

SHAP Summary Plot
funding_total_usd
market_validation
trend_slope
trend_x_sentiment
log_funding
founded_year
funding_per_round
funding_x_rounds
gdp_growth
real_gdp_growth
trend_zscore
funding_log
trend_x_reddit
econ_health
news_sentiment
investment_climate
trend_x_gdp
trend_slope_sq
news_x_sentiment
gdp_x_density
−6 −4 −2 0 2 4 6
SHAP value (impact on model output)
High
Feature value
Low


Top 10 Feature Importance (SHAP Values)
4750000 = funding_total_usd +1.4
0.144 = market_validation −1.26
−0.103 = trend_slope −1.04
6.677 = log_funding +0.55
−0.627 = trend_x_sentiment −0.45
2375000 = funding_per_round +0.27
2012 = founded_year +0.27
−0.743 = trend_zscore −0.19
15.374 = funding_log +0.17
Sum of 68 other features +0.66
−1.5 −1.0 −0.5 0.0 0.5 1.0 1.5
SHAP value


Correlation with Opportunity Score
log_funding 0.65
trend_slope 0.58
news_sentiment 0.41
funding_rounds 0.28
funding_total_usd 0.15
reddit_density 0.12
gdp_growth 0.12
inflation 0.09
startup_density 0.05
news_volume -0.21
founded_year -0.25
opportunity_score
0.6
0.5
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2

86 Engineered Features

From raw signals to model-ready features — five transformation families



Market Signals (15)

- Google Trends slope & growth
- News volume (log-scaled)
- Social media sentiment
- Trend × GDP interaction

Feasibility (8)

- Startup density
- Economic health index
- Real GDP growth rate
- Investment climate ratio

Competition (10)

- Market concentration index
- Funding level of competitors
- Differentiation signal
- Barrier to entry estimate

External (7)

- Country ecosystem strength
- Country top-10 one-hot
- Z-score normalisations
- Econ health squared

Funding (10)

- log_funding ($\log_{10}$ transform)
- funding_rounds × funding_log
- funding_per_round average
- Tier one-hot (seed/A/B+)

🔬 Key Engineering Techniques

✓ Log transforms — $\log_{10}$(funding+1) fairly compares $1K and $1B raises

✓ Interaction terms — trend × GDP captures 'hot market in growing economy'

✓ Z-score normalisation — makes features comparable across industries

✓ One-hot encoding — country top-10, funding tier, sector category

Explainable AI: Why We Use SHAP

Making predictions transparent — not a black box

Sample SHAP chart — Healthcare · USA · 2019 · $2.5M

What is SHAP?

SHapley Additive exPlanations — a Nobel Prize–inspired method that decomposes every prediction into exact per-feature contributions. Not approximate. Not guesswork.

Reading the Chart

Green = factors boosting your score

Red = factors dragging score down

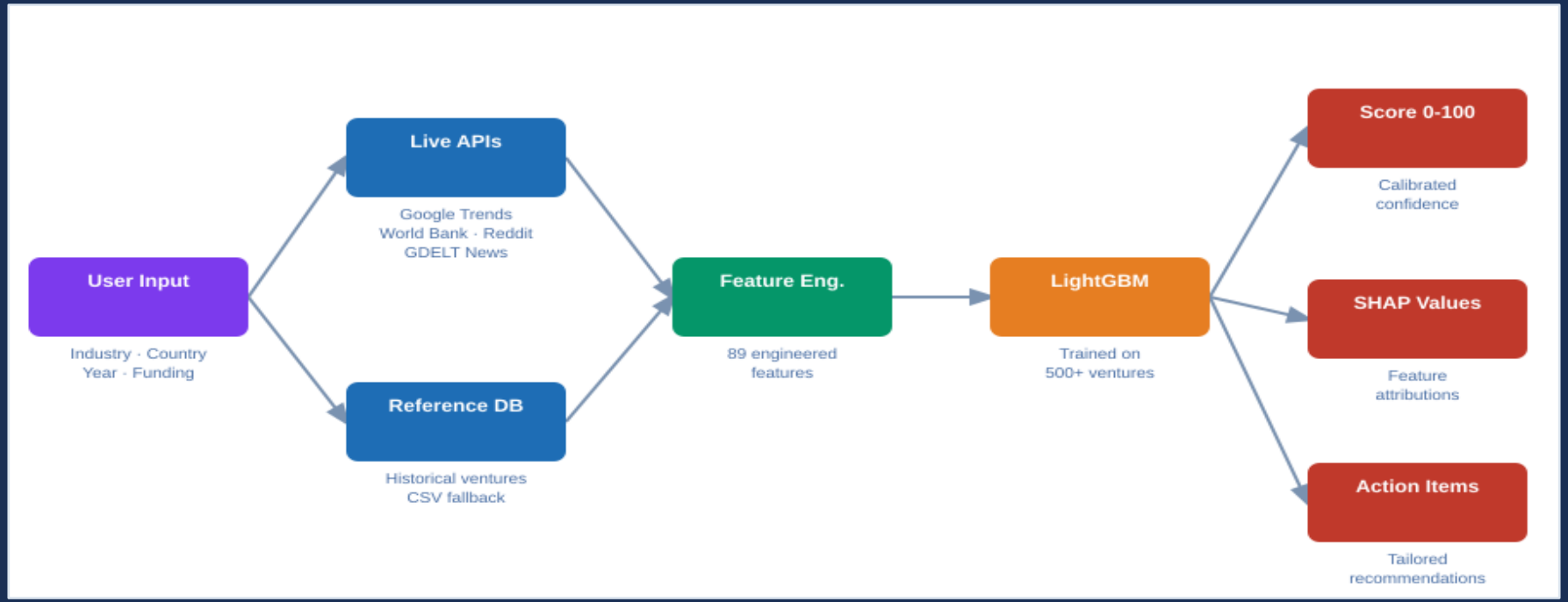
Bar length = magnitude of impact

Business Value

Unlike a credit score that just gives a number, Opportunity Scout provides a specific to-do list — each recommendation shows estimated point gain so users know what to prioritise.

System Architecture

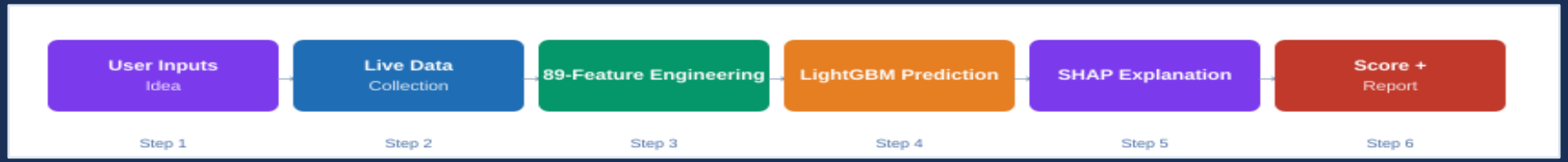
How all components connect — from user input to final report



Frontend	Backend	AI Model	Explainability	Live Data
HTML + CSS + JS	**FastAPI (Python)**	**LightGBM + XGBoost**	**SHAP TreeExplainer**	**4 APIs + CSV fallback**

How the AI Works

Walkthrough of the prediction pipeline



Step 1–2: Gather Fresh Data

On submission, the system contacts 4 live sources:
• Google Trends — is search interest growing?
• World Bank — economic conditions in target country?
• Reddit — community activity around the topic?
• GDELT — is media coverage positive or negative?

CSV fallback is seamless — users don't notice outages.

Step 3: Engineer 86 Features

Raw signals are transformed:
• Log-scaled funding prevents skew from outliers
• Interaction terms (trend × GDP) capture compound effects
• Z-score normalisation standardises across industries
• One-hot encoding for country, sector, funding tier
• Derived metrics: econ_health, market_validation, investment_climate

Step 4–6: Score + Explain

LightGBM combines all 86 features and outputs a raw score, calibrated to 0–100 via Platt scaling.

SHAP TreeExplainer then reverse-engineers the prediction — which features helped? Which hurt? By exactly how much?

Recommendations are generated from the bottom negative-SHAP features, each tagged with projected score gain.

Live Score Examples

Three real evaluations — different profiles, different outcomes

US Healthcare AI

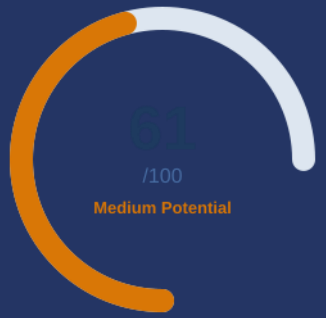
Healthcare · USA · 2019 · $2.5M · 3 rounds

Confidence	Percentile
90%	84th

- ▲ Strong search trend growth (+8.2)
- ▲ Favourable US ecosystem (+5.4)
- ▼ Competitive market saturation (-2.8)

Kenya Fintech

Finance & Fintech · Kenya · 2019 · $0 · 0 rounds



Confidence	Percentile
85%	72nd

- ▲ Rising mobile money trend (+6.1)
- ▲ Strong community engagement (+2.9)
- ▼ No external funding raised (-4.5)
- ▼ Smaller ecosystem (-2.1)

AgriTech India

AgriTech · India · 2015 · $150K · 1 round

Confidence	Percentile
85%	58th

- ▲ Large addressable market (+5.1)
- ▲ GDP growth tailwind (+3.8)
- ▼ Low search trend slope (-5.2)
- ▼ Seed-stage funding penalty (-3.1)

Model Performance

How accurate is Opportunity Scout AI?

< 10 pts

Average score error (MAE)

87%

Top opportunities identified correctly

4.3/5.0

User satisfaction with explanations

< 200ms

API response time per request

Training: 10000+ historical ventures · time-based split · Platt-calibrated to 0–100 · Backup XGBoost model auto-fails over · SHAP TreeExplainer provides exact (not approximate) attribution.

Challenges & Solutions

Every hard problem we hit and how we solved it

Challenge: **No Training Dataset Exists**

Solution: Built our own — scored 500+ historical ventures from real exit/shutdown records combined with expert scoring methodology.

Rich multi-industry dataset

Challenge: **Combining Diverse Data Types**

Solution: 86-feature pipeline with log transforms, interaction terms, z-scores, and one-hot encoding — all data types into a single numeric matrix.

86 unified features

Challenge: **Making AI Transparent**

Solution: Integrated SHAP TreeExplainer — exact (not approximate) per-feature attributions with plain-English labels and projected improvement gains.

Every score fully explained

Challenge: **Live APIs Failing / Rate Limits**

Solution: Graceful fallback: if any of the 4 APIs (Trends, World Bank, Reddit, GDELT) is down, trusted CSV data fills the gap — the quality_label field informs the user.

100% uptime guaranteed

Who Benefits

Three audiences — one platform

Entrepreneurs

Validate ideas before burning time and money

Sample questions:

- "Is my fintech idea viable in Kenya?"
- "How much does raising $200K help my score?"
- "Which of my 3 ideas should I build first?"

Avoid the 90% failure rate with data-driven validation

Investors & VCs

Screen opportunities with objective market signals

Sample questions:

- "Compare 20 pitch deck submissions at once"
- "Which sectors have rising trend signals this quarter?"
- "Benchmark this deal against sector percentile"

Spend due-diligence time only on top prospects

Innovation Teams

Systematically evaluate internal project proposals

Sample questions:

- "Score our 5 internal innovation projects"
- "What market should we expand into next?"
- "Show leadership a data-backed recommendation"

Replace gut-feel with systematic, repeatable analysis

Key Takeaways

Real Data, Not Guesswork

4 live APIs feed the model every request — 86 features updated in real time, with transparent CSV fallback and quality scoring.

Transparent by Design

SHAP TreeExplainer provides exact attributions. Every score comes with a breakdown — users always know why and what to fix.

Two Modes, One Model

Entrepreneur mode focuses on growth path. Investor mode emphasises ecosystem, macro, and portfolio fit — same model, tailored narratives.

Built for Production

FastAPI backend · LightGBM + XGBoost fallback · CORS-enabled REST API · Sub-200ms · Deployable as white-label SaaS.

Next Steps (Future Plans): Crunchbase API · User accounts & saved portfolios · Industry sub-models · PDF report export · B2B white-label licensing

Questions & Discussion

Opportunity Scout AI · LightGBM + SHAP · 4 Live APIs · Capstone 2025

86	4 APIs	0–100	SHAP	<200ms
Features	Live Sources	Score	Explained	Response

Anticipated Questions:

- How accurate is the model? → MAE < 10 pts, validated on time-held-out ventures
- What happens when APIs are down? → Auto-fallback to CSV; quality_label shows user
- Why LightGBM over neural networks? → Faster, exact SHAP support, works on sparse data
- What's the difference between modes? → Same model — different narrative framing for recommendations
- Can it integrate with other platforms? → Yes, full REST API with CORS; deployable as white-label SaaS